

Omer Hayat

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EDUCATION

New York University Abu Dhabi

B.S. in Computer Science

Minors: Applied Mathematics, Philosophy, Artificial Intelligence

Abu Dhabi, UAE

Expected May 2028

EXPERIENCE

Robotics and Machine Learning Research Assistant

Aug. 2025 – Present

iCAS Lab, New York University Abu Dhabi

Abu Dhabi, UAE

- Designed a resource-aware semantic-navigation architecture for the Unitree Go2, separating perception, semantic memory, natural-language querying, safety validation, and ROS 2 navigation into independently testable modules.
- Built a semantic-memory and query engine that resolves natural-language targets, handles ambiguous, unavailable, and unsafe destinations, and publishes schema-validated map-frame goals for Nav2.
- Integrated the ROS 2 packages into the lab's shared navigation stack; independent physical-robot testing achieved over 90% success across about 25 spoken-command trials, including navigation to chair and bag targets.

AdoptAI Fellow

May 2026 – Jul. 2026

AdoptAI, The Lombard Hill Group

Remote

- Built a reusable AI-adoption and economic-assessment framework across 12 Excel and Google Sheets worksheets, using formulas and scripts to standardize three-year ROI, NPV, payback, break-even, and benefit-cost analyses for client engagements.

Software Engineer Intern

Feb. 2025 – Jun. 2025

Enrichly, Techstars '23

Abu Dhabi, UAE

- Owned frontend engineering for an early production Next.js chatbot prototype in a three-intern team, building three core product flows and reusable UI foundations from scratch.
- Replaced a standalone chatbot tab with a site-wide floating assistant and voice-chat interface; integrated a Hugging Face model and built a 15-scenario evaluation harness that surfaced DSM-5 grounding errors and hallucinations, informing prompt, safeguard, and product-flow revisions.

Data Analyst

Jul. 2023 – Aug. 2023

CardPay, NIC LUMS

Lahore, Pakistan

- Built a Python web-data extraction, SQL-cleaning, and Tableau dashboard workflow to rank about 50 SMEs by geography, company size, and customer base, replacing manual prospect research; 6 of the top 10 recommendations became clients during the internship.

PROJECTS

FailSpec | TypeScript, React, Express.js, Playwright, OpenAI Codex

OpenAI Build Week 2026

- Led platform architecture and integration in a three-person team for a local-first developer tool that converts bug reports into evidence-backed Playwright regression tests using OpenAI Codex.
- Owned the investigation API, asynchronous lifecycle orchestration, persistence, and runtime composition, integrating Codex analysis, disposable Git worktrees, controlled Playwright execution, cleanup, and deterministic verification.
- Hardened Windows and POSIX execution with typed failure states, structured-output validation, and sanitized evidence; validated the prototype with 286 primarily hand-written automated tests.

RepoTriage | Python, FastAPI, SQLAlchemy, PostgreSQL, scikit-learn, Docker

- Built a FastAPI/PostgreSQL issue-intelligence service from 1,119 historical pandas issues, exposing APIs for 15-label prediction, confidence-based abstention, similar-issue retrieval, and maintainer feedback.
- Developed leakage-safe training and serving pipelines with chronological splits and a train-only TF-IDF index, achieving 0.942 test recall@10 with calibrated thresholds, versioned artifacts, SQLAlchemy persistence, and Dockerized deployment.

PitchMirror | Next.js, Fastify, FFmpeg, AWS, Amazon Bedrock

Amazon Nova AI Hackathon 2026

- Built and deployed a solo multimodal presentation coach, reducing visual input from thousands of video frames to five representative keyframes while generating structured feedback on voice, presence, and content.
- Engineered an asynchronous AWS workflow across S3, DynamoDB, Step Functions, ECS Fargate, Transcribe, and Bedrock, with idempotent job execution, schema validation, automated JSON repair, and deterministic partial-report fallbacks.

TECHNICAL SKILLS & ACTIVITIES

Languages: Python, TypeScript/JavaScript, C++, SQL

Software Engineering: React, Next.js, Vite, Node.js, Express.js, Fastify, FastAPI, Zod, Playwright, Vitest, pytest

Data & Machine Learning: PostgreSQL, SQLAlchemy, pandas, NumPy, scikit-learn, PyTorch, Hugging Face

Cloud & Systems: AWS (S3, DynamoDB, Step Functions, ECS Fargate, Bedrock), Docker, Linux, ROS 2, Git, FFmpeg

Programs & Certifications: AWS AI & ML Scholars – Future AWS AI Programmer Nanodegree Scholar;

DeepLearning.AI Machine Learning Specialization; Google Data Analytics Professional Certificate

Technical Writing: Founded and grew *The Epoch*, a biweekly ML and software engineering newsletter, to 500+ subscribers in its first month